

Design and Analysis of a Decoupling Current Controller for Induction Machine

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Abstract—This paper analyzes the characteristics of current controllers for vector-controlled AC motors. In synchronous reference frame, the existence of cross-coupling, which increases with the synchronous frequency, impairs the behavior of torque response. System vibration and instability can occur if the current controller does not properly compensate the coupling and delays between the sampling of signals and PWM output. To overcome these problems, a decoupling current control scheme is proposed based on a more accurate induction machine model. The time-delay and corresponding compensation strategy are also studied in this paper. Simulation and experimental results are provided to show the improved dynamic performance of the proposed current controller.

Keywords— current control; decouple; complex vector; Induction motor; vector control

I. INTRODUCTION

Accurate and rapid speed and torque control is an essential requirement for high performance vector-controlled motor drive system. The objective of the vector control is to independently control the torque and flux, so that better dynamic performance, similar to that in a DC motor, is achieved [1]. However, caused by the reference frame transformation, cross-coupling between d-q current appears, which is proportional to the synchronous frequency. As a result, the dynamic characteristics of current controller degrade as the synchronous frequency increases [2].

There have been several strategies proposed for ac machine current control. The synchronous frame PI current regulator has been acknowledged as the standard for current regulation of ac machines, due to its ability of regulating ac signals over a wide frequency range [3], [4]. However, the existence of cross coupling deteriorates the dynamic current response [5]. Reference [6] is a well-known decoupling strategy, which adds the negative quantities of coupling voltages directly to the controller output in a feedforward manner. But its performance is influenced by the parameter sensitivity and its ability is reduced at low switching frequency.

Harnefors and Nee constructed a robust decoupling current controller by using the internal model control (IMC) method [7]. Briz *et al.* proposed a similar current controller, by utilizing the zero-pole cancellation principle, in a way of complex vector notation [8]. Based on the analysis of above

methods, Jung and Nam derived a dynamic decoupling scheme by employing two additional PI controllers in coupling paths [9]. However, these current controllers do not perform satisfactorily, especially operating at very low switching frequency. The reason is that the drive motor and the inverter of these methods are inaccurately modeled [10]. In addition, the high power AC drive system introduces considerable signal delays, which will increase the coupling and affect the behavior of the current control.

This paper uses complex vector variables to establish more accurate models of the machine and the inverter. A decoupling current control method, including analyses of the signal delay and its compensation, is derived. Simulation and experimental results show the improved dynamic performance, stability, and robustness of the current controller.

II. INDUCTION MACHINE MODELS

The electromagnetic system of an induction machine can be described using complex vector notation in the stationary frame, with the stator current and the rotor flux as the state variables:

$$\mathbf{u}_s^s = (R_s + \frac{L_m^2}{L_r \tau_r}) \mathbf{i}_s^s + \sigma L_s s \mathbf{i}_s^s - \frac{L_m}{L_r \tau_r} \Psi_r^s + j \frac{L_m}{L_r} \omega_r \Psi_r^s \quad (1)$$

$$\Psi_r^s = L_m \mathbf{i}_s^s - \tau_r s \Psi_r^s + j \tau_r \omega_r \Psi_r^s \quad (2)$$

Where $\mathbf{u}_s^s$, $\mathbf{i}_s^s$, Ψ_r^s are the stationary frame complex vector notations, total leakage factor is $\sigma = 1 - L_m^2 / (L_s L_r)$; $\tau_r = L_r / R_r$ is the rotor time constant; L_s , L_r , L_m are stator, rotor and mutual inductances; R_s , R_r are stator and rotor resistances; ω_r is rotor angular frequency; s is the derivative operator. The form of stationary frame complex vector is defined as $\mathbf{x}^s = x_\alpha^s + j x_\beta^s$.

Transformed into a synchronous reference frame, the complex vector equations of induction motor are obtained

$$\mathbf{u}_s^e = (R_s + j \sigma L_s \omega_e) \mathbf{i}_s^e + \sigma L_s s \mathbf{i}_s^e + \frac{L_m}{L_r} s \Psi_r^e + j \frac{L_m}{L_r} \omega_e \Psi_r^e \quad (3)$$

$$\Psi_r^e = L_m \mathbf{i}_s^e - \tau_r s \Psi_r^e - j \tau_r \omega_{sl} \Psi_r^e \quad (4)$$

Where the slip frequency is $\omega_{sl} = \omega_e - \omega_r$, ω_e is the synchronous angular frequency. The form of synchronous frame complex vector is defined as $\mathbf{x}^e = x_d^e + jx_q^e$. Compare (3) with the converted scalar form of it, we can conclude that the stator voltage contains two parts, the cross-coupling term $j\sigma L_s \omega_e \mathbf{i}_s^e$ and the back-EMF voltage $jL_m / L_r \omega_e \Psi_r^e$, which are both dependent on the synchronous frequency ω_e , resulting a deterioration of the dynamic response as the synchronous frequency increases. If these two terms are decoupled, the mutual coupling between the flux and torque will disappear, so will the dependence on the synchronous frequency. Through simultaneous equations (3) and (4), the representation of the stator voltage can be expressed as

$$\begin{aligned} \mathbf{u}_s &= (R_s + \sigma L_s s + j\sigma L_s \omega_e + \frac{(L_m / L_r)^2 R_r (s + j\omega_e)}{s + R_r / L_r + j\omega_{sl}}) \mathbf{i}_s \\ &= (R_s + \sigma L_s s + j\sigma L_s \omega_e + \frac{b + jc}{a}) \mathbf{i}_s \end{aligned} \quad (5)$$

Where the variables of the above equation are $b/a = (L_m / L_r)^2 R_r (s^2 + sR_r / L_r + \omega_e \omega_{sl}) / ((s + R_r / L_r)^2 - \omega_{sl}^2)$, $c/a = (L_m / L_r)^2 R_r (\omega_e s + \omega_e R_r / L_r) / ((s + R_r / L_r)^2 - \omega_{sl}^2)$. Corresponding to the complex vector equation (5), the equivalent scalar voltage equations (6) and (7) can be derived, which are based on the accurate induction motor model.

$$u_{sd} = (R_s + \sigma L_s s + b/a) i_{sd} - (c/a + \sigma L_s \omega_e) i_{sq} \quad (6)$$

$$u_{sq} = (R_s + \sigma L_s s + b/a) i_{sq} + (c/a + \sigma L_s \omega_e) i_{sd} \quad (7)$$

The decoupling methods of reference [4]-[9] are mostly based on the simplified induction motor model of the RL load, by considering the back-EMF term as a disturbance to the stator voltage equation. In that case, the two components of the machine model b/a and c/a are neglected. However, the back-EMF term will vary proportional to the rotor speed, which means it is very dependent on the inertia and acceleration properties of the motor and load. For modern servo drives, these dynamics are often as fast as electrical dynamics and can play a significant role [5]. In addition, the back-EMF voltage also contains a coupling part in the stator voltage, the proportion of which in (5) will grow larger as the speed increases. Thus, in order to realize the perfect decoupling, it is essential to establish a complete machine model like (6) and (7).

III. A PROPOSED DECOUPLING CURRENT CONTROLLER

A. Design of the current controller with no time delay

The decoupling current controller is designed to decouple the motor d-axis current from the q-axis current, and to regulate the two currents independently. Synchronous frame current controller makes the variables become dc quantities to maintain zero steady-state error, whose characteristics are superior to the stationary frame PI controller. However, due to the cross-

coupling and the dependence on the synchronous frequency, its dynamic response is far from being ideal.

In this paper, an improved current controller is proposed, and the block diagram of the decoupling scheme is shown in Fig. 1. The right side of the dotted line is the induction machine model, the left side is the designed current controller, and the decoupling terms G_3 and G_4 are introduced behind the difference of command current values and the feedback values. Then, we obtain from the principle diagram Fig. 1 that

$$\begin{aligned} G_1(i_{sd}^* - i_{sd}) + G_4(i_{sq}^* - i_{sq}) &= (R_s + \sigma L_s s + b/a) i_{sd} \\ &- (c/a + \sigma L_s \omega_e) i_{sq} \end{aligned} \quad (8)$$

$$\begin{aligned} G_2(i_{sq}^* - i_{sq}) + G_3(i_{sd}^* - i_{sd}) &= (R_s + \sigma L_s s + b/a) i_{sq} \\ &+ (c/a + \sigma L_s \omega_e) i_{sd} \end{aligned} \quad (9)$$

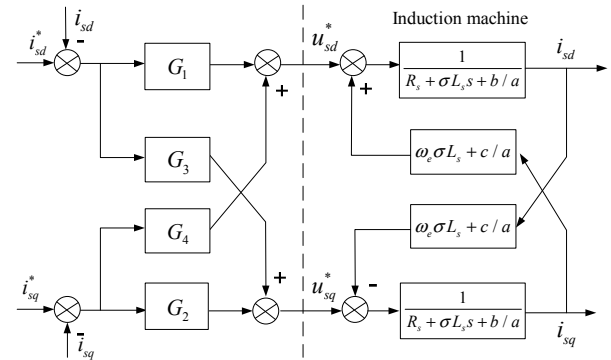


Figure 1. The principle diagram of the decoupling current controller

Simplifying the two equations, the following expression can be achieved

$$\begin{aligned} &[(R_s + \sigma L_s s + b/a)^2 + G_1 G_2 + (G_1 + G_2)(R_s + \sigma L_s s + b/a) \\ &+ (c/a + \sigma L_s \omega_e)^2 - G_3 G_4 + (G_3 - G_4)(c/a + \sigma L_s \omega_e)] i_{sd}^* \\ &= [G_1(R_s + \sigma L_s s + b/a + G_2) + G_3(-G_4 + c/a + \sigma L_s \omega_e)] i_{sd}^* \\ &+ [G_4(R_s + \sigma L_s s + b/a + G_2) + G_2(-G_4 + c/a + \sigma L_s \omega_e)] i_{sq}^* \end{aligned}$$

To cancel out the coupling term, the d-axis current i_{sd} is supposed to be controlled by its command value i_{sd}^* and independent from i_{sq}^* , which means the coefficient of i_{sq}^* ought to be zero. So the expression (10) is obtained, and (11) can be got in the same way.

$$G_4(R_s + \sigma L_s s + \frac{b}{a} + G_2) + G_2(-G_4 + \frac{c}{a} + \sigma L_s \omega_e) = 0 \quad (10)$$

$$G_3(R_s + \sigma L_s s + \frac{b}{a} + G_1) - G_1(G_3 + \frac{c}{a} + \sigma L_s \omega_e) = 0 \quad (11)$$

Finally the decoupling terms G_3 and G_4 are derived

$$G_3 = \frac{c/a + \sigma L_s \omega_e}{R_s + \sigma L_s s + b/a} G_1 \quad (12)$$

$$G_4 = -\frac{c/a + \sigma L_s \omega_e}{R_s + \sigma L_s s + b/a} G_2 \quad (13)$$

G_1 and G_2 are supposed to be functioned as PI regulators, and the outputs of two axes are the corresponding non-coupling voltage term. But as to the complete machine model, the zeros of G_1 and G_2 should make it possible to cancel the poles of the non-coupling term. Thus, $G_1 = G_2 = \omega_{cb}(R_s + \sigma L_s s + b/a)/s$, where ω_{cb} is the desired bandwidth of the current loop.

Owing to the symmetric form of the two-axis machine model and proposed controller, we can use the complex vector transfer function to analyze the performance of controller. Thus, the open-loop complex vector transfer function of the current controller can be given by

$$G_r(s) = G_1 + j(c/a + \sigma L_s \omega_e)G_1 / (R_s + \sigma L_s s + b/a) \quad (14)$$

$$= \omega_{cb} [R_s + \sigma L_s s + b/a + j(c/a + \sigma L_s \omega_e)] / s$$

According to (5), we have got the transfer function of the machine model

$$G_m(s) = I_s(s) / U_s^*(s) = 1 / [R_s + \sigma L_s s + b/a + j(c/a + \sigma L_s \omega_e)] \quad (15)$$

Combining (14) and (15), the closed-loop transfer function describing the system is derived as follows.

$$G_{clo}(s) = \frac{I_s(s)}{I_s^*(s)} = \frac{G_r(s)G_m(s)}{1 + G_r(s)G_m(s)} = \frac{\omega_{cb}}{s + \omega_{cb}} \quad (16)$$

It is noted that, there are poles introduced by the cross coupling, imaginary contents of machine model (15), which can vary with the synchronous frequency. In terms of this, our proposed current controller is able to make the cancellation of the poles and zeros, enabling fast and accurate current regulation. Through the analysis of complex vector transfer functions, the result of the proposed controller is similar with reference [11], demonstrating the validity of this approach from another perspective.

B. Time delay and its compensation scheme

Time delay is the intrinsic problem of full-digital implementation, which is related to the time sequence shown in Fig.2. The time sequence diagram contains sampling of stator current, calculation of voltage command values, PWM update of the underflow interrupt, and PWM output, where T_s is the sampling period. One sampling time is delayed during the execution of the control algorithm, which means $T_{cal} = T_s$. Because the average output voltage of the inverter is supposed to change linearly, there will be a time delay between the sampling time $t + T_s$ and $t + 2T_s$, which is generally obtained $T_{PWM} = 0.5T_s$. Thus, the overall time delay including the signal processing time and PWM output time is approximately $T_d = T_{cal} + T_{PWM} = 1.5T_s$.

The sampling delay is commonly represented by a first-order system $G_s(s) \approx 1/(T_d s + 1)$. When transformed into synchronous coordinates, it is shown as

$$G_d^e = \frac{1}{T_d s + 1 + j\omega_e T_d} \quad (17)$$

In the drive system at high switching frequency, the inverter and sampling delay is negligible. But in high power drive system, phase and magnitude errors introduced by time delay deteriorate system performances considerably, so it is essential to study the effect the delay and make a compensation of it.

In this paper, to reduce the adverse effect of time delay, we design a varying zero to cancel the pole of the delay, and a first-order model to avoid the disturbance of differentiation element. The expression of the decoupling current controller containing delay compensation is shown in (18), and the diagram of the controller is given by Fig.3.

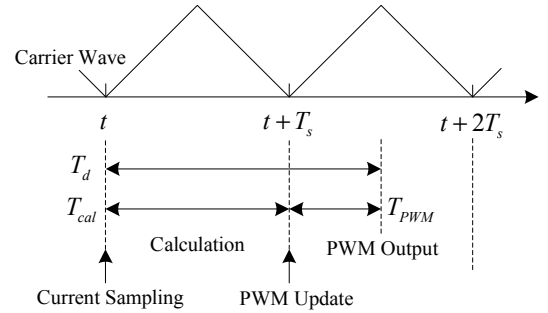


Figure 2. Typical time sequence of digital control

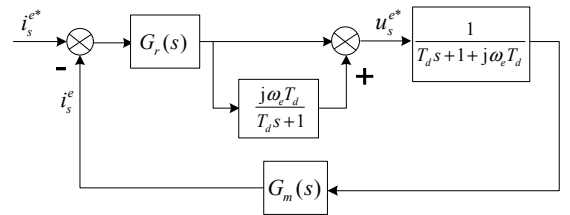


Figure 3. Diagram of decoupling current control

$$G_{cv}(s) = \omega_{cb} \frac{R_s + \sigma L_s s + \frac{b}{a} + j(\frac{c}{a} + \sigma L_s \omega_e)}{s} \cdot \frac{T_d s + 1 + j\omega_e T_d}{T_d s + 1} \quad (18)$$

C. Performance of the proposed controller

Switching frequency of the inverter has a significant influence on the performance of vector-controlled drive system. In high power and low switching frequency system, the dynamic decoupling characteristic is not as good as we expect. The analysis of this paper is mainly based on this situation. The pole-zero map of synchronous PI controller is shown in Fig.4, where the switching frequency is 500Hz, the rated frequency of the motor is 50Hz, and the rotor frequency is varying from 0Hz to 100Hz. On one hand, the classical PI controller has only constant zeros, and the varying poles can't be cancelled as the synchronous frequency increases. Meanwhile, there is one

dominated pole P_1 moving forward to the imaginary axis, which may lead to system instabilities. On the other hand, the existence of time delay may increase the cross coupling, shown in Fig.4 of pole P_2 and P_3 .

Figure 4. Pole-Zero maps of the system with the change of speed using synchronous PI control

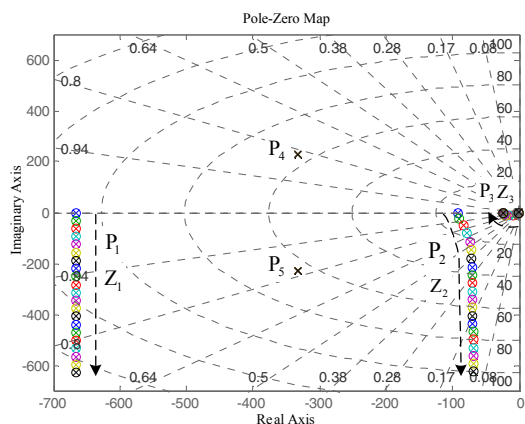


Figure 5. Pole-Zero maps as speed varies using the new controller

the new method can keep a nice decoupling and stable property compared to the classical PI controller.

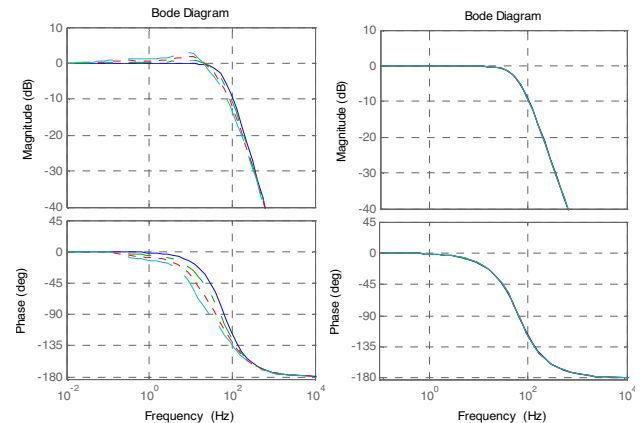


Figure 6. The closed-loop frequency characteristic of current loop

Figure 7. Closed-loop amplitude frequency response of parameter error

The dynamic performances of classical PI controller and the proposed decoupling current controller are evaluated and compared using both simulation and experiments, with same system parameters. The induction motor has the following parameter values: $R_s = 1.539\Omega$, $R_r = 0.519\Omega$, $L_{ls} = 11.27\text{mH}$, $L_m = 464.9\text{mH}$, power rating=7.5kW, frequency rating=50Hz, switching frequency $f_c = 500\text{Hz}$, and four poles.

shows experimental results of the new method with the given torque step from 10Nm to 30Nm, where the system is based on TMS320F28335 Digital Signal Processor. In spite of the deviation of motor parameters, the response of d , q -axes current only has a small disturbance, which validates the properties of previous analyses. In Fig.10 the dynamic response of two different methods with different rotor frequency is given. Note from comparison of Fig.10 (a) and (b), the classical PI controller shows increased couplings and decreased dynamics as speed rises, while the proposed method of Fig.10(c) and (d) displays better performances. When contrasted between two different methods with the same speed, Fig.10 (a) and (c) as well as Fig.10 (b) and (d), the proposed controller reveals evident advantage compared to synchronous PI controller.

V. CONCLUSION

This paper has attempted to solve the problem of cross coupling, the incomplete decoupling included, and time delay in the high performance AC vector drive system. A decoupling current controller is derived from complete models of the motor and inverter, along with the time delay compensation strategy, offering a new perspective on decoupling current control issues. The improved decoupling property and dynamic response property, as well as the robustness to parameter errors are confirmed both by theory analysis and experimental results.

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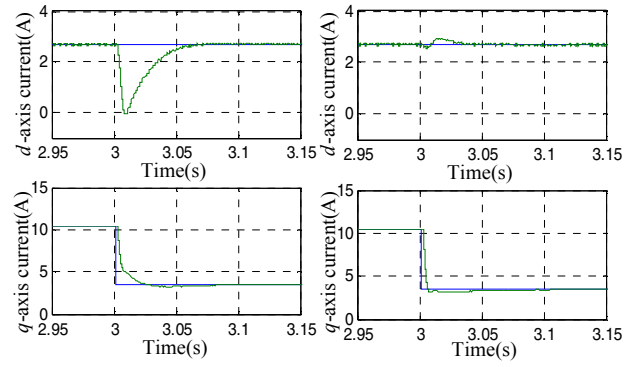


Figure 8. Simulation waveforms of the two current controllers

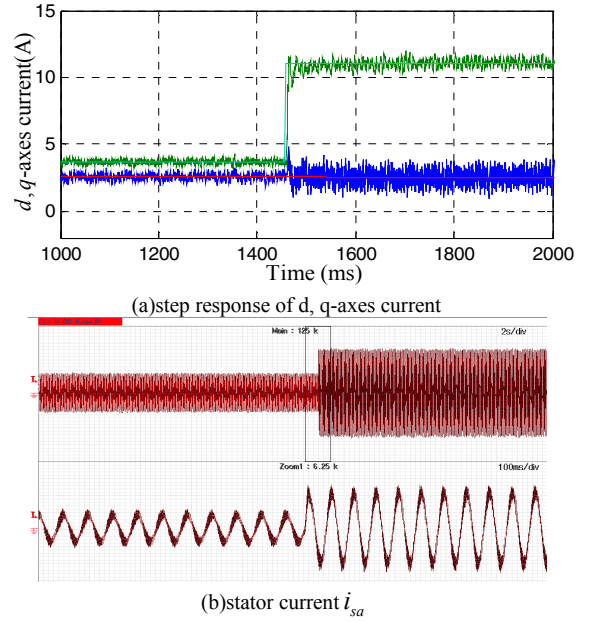


Figure 9. Experimental results of proposed controller

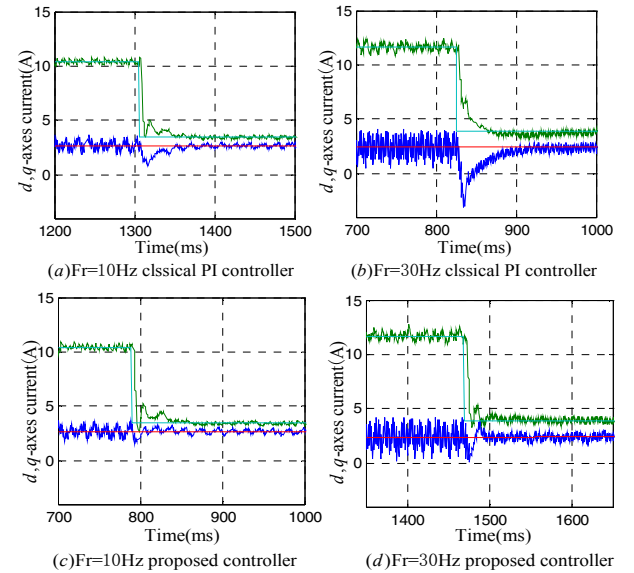


Figure 10. Comparison of two methods in different rotor frequency